

Inferring causal networks for genomic trios with high-dimensional confounding variables

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16 **Abstract**

17 Network inference under the principle of Mendelian randomization (PMR) enables the study
18 of causal relationships between two molecular phenotypes (e.g., gene expression) by lever-
19 aging genetic variants as instrumental variables to impose constraints on edge direction. In
20 practice, inference is often affected by numerous confounding variables, such as other genes,
21 other processes and demographic variables, making it essential to robustly account for con-
22 founders. However, existing methods have limited ability to accommodate high-dimensional
23 confounding. Here, we introduce Mendelian Randomization for Genomic Networks (MRGN),
24 a regression-based method under the PMR for inferring causal relationships in genomic trios
25 from individual-level data while accounting for many confounders. MRGN extends beyond
26 the canonical mediation model to distinguish among multiple causal relationships. By prior-
27 itizing conditional independence tests given the genetic variant and confounders, MRGN can
28 detect causal relationships between phenotypes even when their marginal association is weak.
29 Through extensive simulations, we demonstrate improved inference accuracy, robustness, and
30 computational efficiency over existing approaches. We further apply MRGN to data from the
31 Genotype-Tissue Expression (GTEx) Consortium to investigate how genetic variants may reg-
32 ulate gene expression in cis and trans.

33 **Keywords**

34 principle of Mendelian randomization, causal network inference, confounding, gene regulatory
35 networks

36 **Key Points**

- 37 • Mendelian Randomization for Genomic Networks (MRGN) is a regression-based method
38 for inferring causal relationships in genomic trios from individual-level data while ac-
39 counting for many confounders. A genomic trio consists of two molecular phenotypes
40 and a genetic variant.
- 41 • MRGN leverages the Principle of Mendelian randomization (PMR), which uses genetic
42 variants as instrumental variables to impose constraints on edge directions in causal net-
43 work inference.
- 44 • By prioritizing conditional independence tests given the genetic variant and confounders,
45 MRGN can detect causal relationships between phenotypes even when their marginal
46 association is weak.

47 **1 Introduction**

48 An important goal in biology is to understand how genes regulate one another and how genetic
49 variation influences gene expression [1, 2, 3, 4]. Gene regulation involves complex networks of
50 high dimensions, which may be represented by directed networks where directed edges repre-
51 sent causal (or regulatory) relationships. Although the ultimate goal in biology is to reconstruct
52 gene regulatory networks at the genome level, it is often of interest to understand the causal re-
53 lationship between a molecular phenotype (e.g. the expression level) of two genes, or between
54 two molecular phenotypes (e.g., between gene expression and DNA methylation), which may
55 be influenced by many other variables such as other genes, other biological processes, demo-
56 graphic variables, etc.

57 To learn the (directed) causal relationship between two phenotypes, we can apply the prin-
58 ciple of Mendelian randomization (PMR) and leverage the information in genetic variants. The

59 PMR assumes that genotypes in a natural population are randomly assigned, thus observational
60 omics data can be viewed as being collected from a randomized experiment [5, 6]. This implies
61 that an edge connecting a genotype to a phenotype will always point to the phenotype, but not
62 the other way around [7, 8]. As a result, this constraint on edge direction makes it possible to
63 break the symmetry of association and draw inferences about the regulatory structure among
64 phenotypes [5, 7, 9].

65 Methods based on the PMR framework differ fundamentally from classical Mendelian ran-
66 domization (MR) methods. Traditional MR focuses on estimating causal effects and depends
67 on three key assumptions [5, 10, 11]: (i) relevance, where the genetic variant is associated with
68 the molecular phenotype; (ii) independence, where the variant is independent of unmeasured
69 confounders due to random allele segregation; and (iii) exclusion restriction, where the variant
70 influences downstream phenotypes only through the intermediate phenotype, and not through
71 other pathways. In contrast, our PMR-based framework does not require these assumptions to
72 hold in full, nor is causal effect estimation its primary objective. Instead, following our earlier
73 work [7, 12, 13], we use the PMR as a structural constraint that limits edge directions in the
74 network. Specifically, edges from molecular phenotypes to genetic variants are not allowed in
75 the network.

76 Many methods have been developed under assumptions similar to the PMR [14, 15, 16,
77 17, 7, 18, 19, 20], although some of these methods are interested mostly in the mediation
78 model $V \rightarrow T_1 \rightarrow T_2$, where V is the genetic variant, T_1 and T_2 are the variables of interest.
79 In this model, the mediator T_1 relays the regulatory information from V to the outcome T_2 ;
80 for example, a genetic variant regulates the expression level of a transcription factor, and its
81 reduced expression leads to its downstream target gene being under-expressed. However, other
82 causal relationships may exist among the three variables.

83 We previously developed MRPC, a method that goes beyond mediation and infers five
84 mutually exclusive, basic causal models for a trio under the PMR (Figure 1) [7, 12, 13]. They

85 include a null model ($V \rightarrow T_1$ and no relationship between T_1 and T_2 , denoted M0) and four
 86 non-null models: mediation (denoted M1), v-structure (or reverse causation; $V \rightarrow T_1 \leftarrow T_2$,
 87 denoted M2), conditional independence (or horizontal pleiotropy; $T_1 \leftarrow V \rightarrow T_2$, denoted
 88 M3), and a fully connected model ($T_1 \leftarrow V \rightarrow T_2$ and $T_1 \leftrightarrow T_2$, denoted M4). The true model
 89 behind M4 may be directed: the edge between T_1 and T_2 is $T_1 \rightarrow T_2$ or $T_1 \leftarrow T_2$, but these
 90 two directions result in the same likelihood for the trio and are therefore Markov equivalent.
 91 Without additional information, they are indistinguishable and only a bidirected/undirected
 92 edge can be inferred for M4.

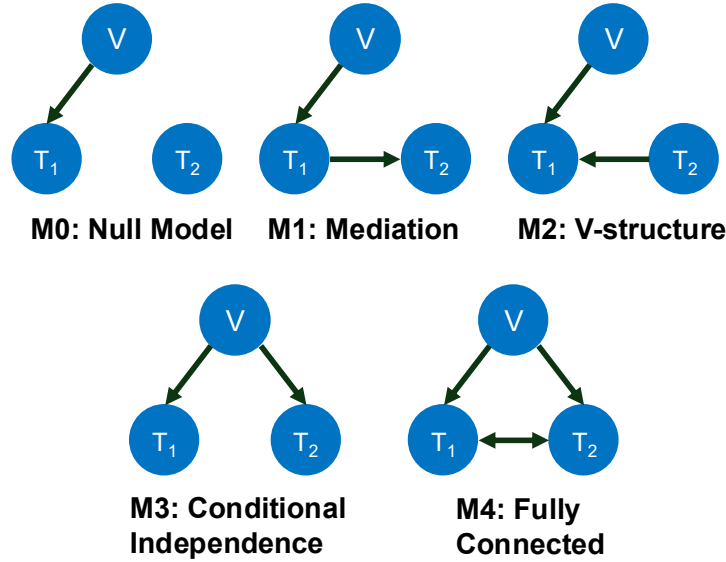


Figure 1: The five basic causal models for a genomic trio that may be inferred under the principle of Mendelian randomization. A genomic trio consists of a genetic variant V and two molecular phenotypes T_1 and T_2 . This figure is adapted from [7].

93 The PMR and the five basic causal models require that no confounding variables can ex-
 94 plain away the directed edge between T_1 and T_2 , since this edge is our main interest. How-
 95 ever, as mentioned earlier, many variables can influence the inference due to the complexity
 96 of gene regulation. A confounding variable may influence the inference in different ways.
 97 Yang et al. [16] proposed three types of confounding C for the $T_1 - T_2$ edge: as a con-
 98 founder ($T_1 \leftarrow C \rightarrow T_2$), an intermediate variable ($T_1 \rightarrow C \rightarrow T_2$), or a common child

99 $(T_1 \rightarrow C \leftarrow T_2)$. Specifically, a confounder is a common parent to both T_1 and T_2 and can
 100 induce a false edge between T_1 and T_2 when unaccounted for. This is similar to T_1 and T_2
 101 being marginally dependent in M3. An intermediate variable acts like a mediator between T_1
 102 and T_2 and can potentially explain away a true $T_1 \rightarrow T_2$ edge when this intermediate variable is
 103 accounted for in the inference, leading to a false negative conclusion. By contrast, a common
 104 child can induce a false edge between T_1 and T_2 when accounted for, leading to a false positive
 105 conclusion. This is because when the common child is accounted for in the inference, T_1 and
 106 T_2 become conditionally dependent, similar to their relationship in M2.

107 Accounting for confounding variables is therefore crucial in causal network inference [21,
 108 16, 22, 23, 24]. Two basic approaches have emerged to address latent confounding between an
 109 exposure and outcome. In the first approach, candidate confounding variables are incorporated
 110 into the graph as additional nodes in the network and also treated in the same way as other
 111 nodes; our MRPC method and several other methods take this approach [7, 23]. However, this
 112 approach is computationally intensive, as the computational complexity for graph inference
 113 grows super exponentially with the number of nodes. The second approach focuses on identi-
 114 fying and accounting for confounders [21, 16, 22]. This approach includes all the confounders
 115 as a whole in the tests for edge presence, and does not increase the size of the network. For
 116 example, we can include the confounders when testing the $T_1 - T_2$ edge in a trio. The trio re-
 117 mains a network of three nodes, but the tests are conditioned on these confounders. However,
 118 these methods focus on the mediation model.

119 Herein we present Mendelian Randomization for Genomic Networks (MRGN), a novel
 120 machine learning method for inferring the causal model of a trio from individual-level data in
 121 the presence of potentially hundreds of confounding variables. MRGN builds on MRPC and
 122 infers the five basic causal models. We take the second approach to confounding described
 123 above and focus on accounting for confounders, and not other types of confounding variables.
 124 This approach allows for the inclusion of many confounding variables all at once, while main-

125 taining high power and fast computation times. We demonstrate the performance of MRGN
 126 and compare to other methods through extensive simulation studies. We further assess this
 127 method in our study of the cis- and trans-regulatory modes of genetic variants, using the whole
 128 blood data from the Genotype-Tissue Expression (GTEx) Consortium [25].

129 **2 Methods**

130 **2.1 The MRGN method**

131 The five causal models for a trio can each be described by a set of conditional and marginal
 132 dependence relations (Figure 2a; Table 1), which may be encoded in the coefficients of two
 133 regression models that account for confounders, assuming that T_1 and T_2 are centered so that
 134 the intercept is zero:

$$T_1 = \beta_{11}V + \beta_{12}T_2 + \mathbf{\Gamma}_1\mathbf{U} + \epsilon_1; \quad (1)$$

$$T_2 = \beta_{21}V + \beta_{22}T_1 + \mathbf{\Gamma}_2\mathbf{U} + \epsilon_2, \quad (2)$$

135 where β s are coefficients of interest, vector $\mathbf{\Gamma}_i$ the coefficients for the confounder vector $\mathbf{U}_i$,
 136 and ϵ_i the random error that follows a normal distribution with mean 0 and standard deviation
 137 σ_i , where $i = 1, 2$.

138 Our conditional independence tests test whether each β is zero, as each β represents the
 139 association given the presence of other variables. For example, in the mediation model of
 140 $V \rightarrow T_1 \rightarrow T_2$, any two variables are dependent on each other, but T_2 and V are conditionally
 141 independent given T_1 . These relations translate to (i) β_{21} being zero, which represents the
 142 association between V and T_2 conditioned on T_1 and the confounders $\mathbf{U}$, and (ii) the other
 143 three β s being nonzero, which represent those associations not being explained away by other
 144 variables. Similarly, models M0 and M3 are uniquely determined by conditional associations,

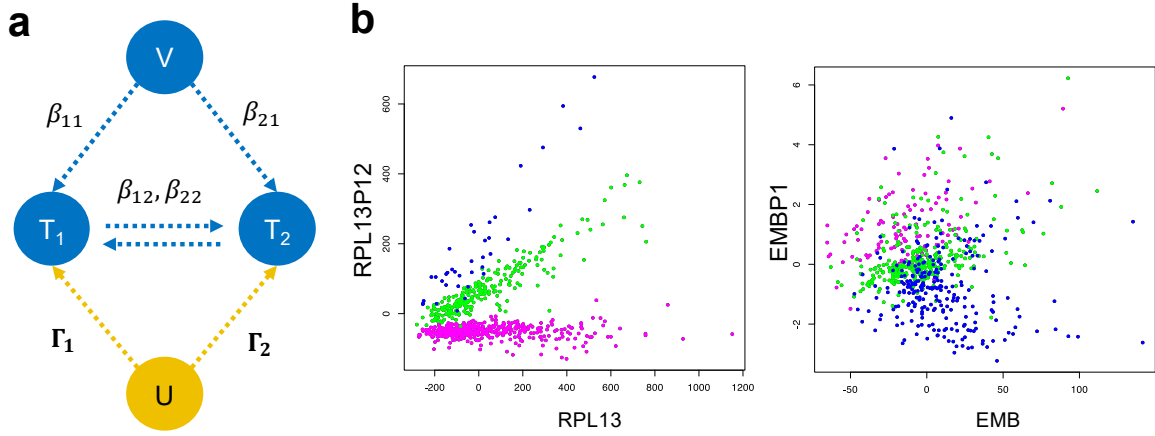


Figure 2: The MRGN method. (a) A genomic trio includes a genetic variant (V), two molecular phenotypes (T_1 and T_2), and a set of confounders (U). (b) Two examples of Simpson's paradox in genomic trios from the GTEx whole blood data. Each dot represents an individual, with colors indicating genotypes. Ignoring genotype, the association is weakened in the left example and disappears in the right example.

for which we only need to conduct conditional independence tests.

In the v-structure model M2, consider the configuration $V \rightarrow T_1 \leftarrow T_2$. As the parents of T_1 , nodes V and T_2 are marginally independent but conditionally dependent given the presence of their child T_1 . Therefore, all the β s, which represent conditional associations, should be nonzero. Similarly, all the β s are also nonzero in the fully connected model M4. The distinction between M2 and M4 lies in the marginal associations: V and T_2 are marginally independent in M2 but are marginally dependent in M4. MRGN therefore performs additional tests on marginal dependence to distinguish these two models.

Note that MRGN prioritizes conditional independence tests because the relationship between T_1 and T_2 within each genotype may weaken or disappear when the genotypes are combined, as in Simpson's Paradox (Figure 2b). This strategy is a substantial improvement over MRPC, which tests marginal independence before conditional independence. n MRPC, a negative marginal test permanently removes the $T_1 - T_2$ edge from further consideration, potentially resulting in low power.

159 MRGN performs the standard t tests on the regression coefficients for conditional depen-
160 dence and the standard test on Pearson correlation for marginal dependence. Since the number
161 of tests for the edges of interest is at most six, we set the significance level for each test to
162 be 0.01, which is equivalent to Bonferroni correction at a familywise type I error rate of 0.05.
163 When the data is not informative enough for any of the basic models to be inferred, MRGN
164 categorizes the trio as “other”.

165 2.2 Confounder selection

166 We perform confounder selection before causal network inference, using a two-step procedure
167 similar to that in the GMAC method [16]. First, we identify candidate confounders from a
168 pool of potential confounding variables. Confounders, which are defined as parents of T_1 and
169 T_2 , should be independent of V . We therefore exclude variables that are marginally associated
170 with V . Next, we select confounders based on their statistical association with either T_1 or T_2 .
171 Similar to [16], we test whether the regression of both T_1 and T_2 on a candidate confounder is
172 statistically significant. Both steps allow for an additional step of the q -value method to control
173 the false discovery rate (FDR) [26].

174 In our R implementation, we substantially improve computational efficiency in two key
175 ways. First, we implemented parallelization in both steps, allowing for faster evaluation when
176 the pool of confounders is large. Second, we used the R package *propagate* for correlation
177 matrix computation by partitioning the problem into smaller, memory-efficient blocks [27],
178 which scales more efficiently to large genomic datasets.

Table 1: Conditional and marginal independence tests in MRGN. Regression coefficients β_{11} and β_{12} (from Equation 1) and β_{21} and β_{22} (from Equation 2) are used to test conditional independence. Marginal independence can be tested through correlations r_{V,T_1} and r_{V,T_2} . The first three models each have two possible configurations.

[illegible]

2.3 Simulation study

Simulation setup

Each simulated trio consists of a bi-allelic genetic variant V with three genotypes and two genes (T_1, T_2) connected under one of the five causal models. Nodes in a trio are generated in topological order. The genotype V_i is simulated first under Hardy–Weinberg equilibrium:

$$\Pr(V_i = 0) = (1 - \theta)^2, \quad \Pr(V_i = 1) = 2\theta(1 - \theta), \quad \Pr(V_i = 2) = \theta^2,$$

where the minor allele frequency θ is sampled uniformly from $(0.01, 0.5)$, such that roughly 10% of trios contain a less common allele with a $< 5\%$ frequency.

Each remaining node Y_j (a gene or a confounder) is sampled from

$$Y_j \sim N\left(\sum_{i \in S_j} \beta_i Y_i, \sigma^2\right),$$

where S_j is the parent set of Y_j , which may include V , the other gene, or confounders. The effect size of V on its target genes is sampled uniformly from $[0.5, 1.5]$ [16]. The effect size between T_1 and T_2 is sampled uniformly from $[0.5, 1]$. Residual variance is set to $\sigma = c\beta$, where β is the T_1 – T_2 effect and c sampled uniformly from $[0.3, 1.5]$.

For M4, the true edge between T_1 and T_2 can be in either direction. For simplicity, we fix all M4 trios to the configuration $T_1 \leftarrow V \rightarrow T_2$ with $T_1 \rightarrow T_2$.

For each trio, we further include three types of confounding variables: a varying number of confounders (between 1 and 50), one intermediate variable, and one common child. Effect sizes for these variables are sampled independently: confounder and intermediate effects from $[0.15, 0.5]$ and common-child effects from $[0.5, 1.5]$, each with a 50% probability of being negative.

We simulate 1,800 trios by sampling from the five causal models with equal probability,

199 and obtain approximately 360 trios per model. For each trio, we draw a distinct combination
 200 of effect sizes, noise level, and minor allele frequency. The sample size is fixed at $n = 670$,
 201 matching the whole-blood sample size of GTEx v8 [25] in our real-data application.

202 **Performance metrics**

203 We evaluated three sets of recall (i.e., power) and precision to quantify the performance on the
 204 entire model of a trio, on individual edges and specifically on the $T_1 - T_2$ edge.

- 205 1. On the causal model type: a trio is considered correctly inferred when its inferred model
 206 type matches its true type, thus requiring all edges to be inferred correctly.

$$\text{Recall}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios of the causal model M}}; \quad (3)$$

$$\text{Precision}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios inferred to be of the causal model M}}. \quad (4)$$

- 207 2. On individual edges: we first calculated recall and precision based on individual edges
 208 in a single trio. Similar to our earlier work [7], we counted an edge with the wrong
 209 direction as a half correct edge.

$$\text{Recall}_{\text{edges}} = \frac{\text{Number of correct edges in inferred graph}}{\text{Total edges in true graph}}; \quad (5)$$

$$\text{Precision}_{\text{edges}} = \frac{\text{Number of correct edges In inferred graph}}{\text{Total edges in inferred graph}}. \quad (6)$$

210 We then calculated the mean metric across all trios.

- 211 3. On the $T_1 - T_2$ edge (allowing for comparison with the GMAC method): the direct of

212 this edge was ignored for this evaluation. Across all trios,

$$\text{Recall}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with a true } T_1 - T_2 \text{ edge}}; \quad (7)$$

$$\text{Precision}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with an inferred } T_1 - T_2 \text{ edge}}. \quad (8)$$

213 3 Results

214 3.1 Simulation Results

215 MRGN improves power and precision to detect edges in the presence of confounding

216 Before applying MRGN, we examined two approaches to confounder selection: controlling the
 217 FDR at 5% with the q -value method (denote “CS- q ”), and not correcting for multiple testing
 218 but instead setting the type I error rate of each individual test at $\alpha < 0.01$ (denoted “CS- α ”).
 219 We evaluated their ability to identify true confounders using precision and recall. Overall, both
 220 methods achieved recall greater than 90% when the number of true confounders per trio was
 221 modest (e.g., fewer than 25; see Figure 3). However, recall under CS- q dropped to approxi-
 222 mately 65% for trios with many (>25) true confounders. As expected, CS- α achieved nearly
 223 perfect recall across all simulations, but with substantially lower precision. Whereas the actual
 224 FDR (i.e., 1-precision) under CS- α was above 70% and increased to above 90% as the number
 225 of true confounders decreased, the actual FDR under CS- q was generally around 20%, already
 226 much higher than the expected level, and increased to 50% as the number of true confounders
 227 decreased. This likely reflects dependence among tests involving the same or highly correlated
 228 variables.

229 MRGN produced qualitatively similar inference results with both confounder selection
 230 methods (Table 2). Both versions achieved substantially higher recall for the mediation, v-

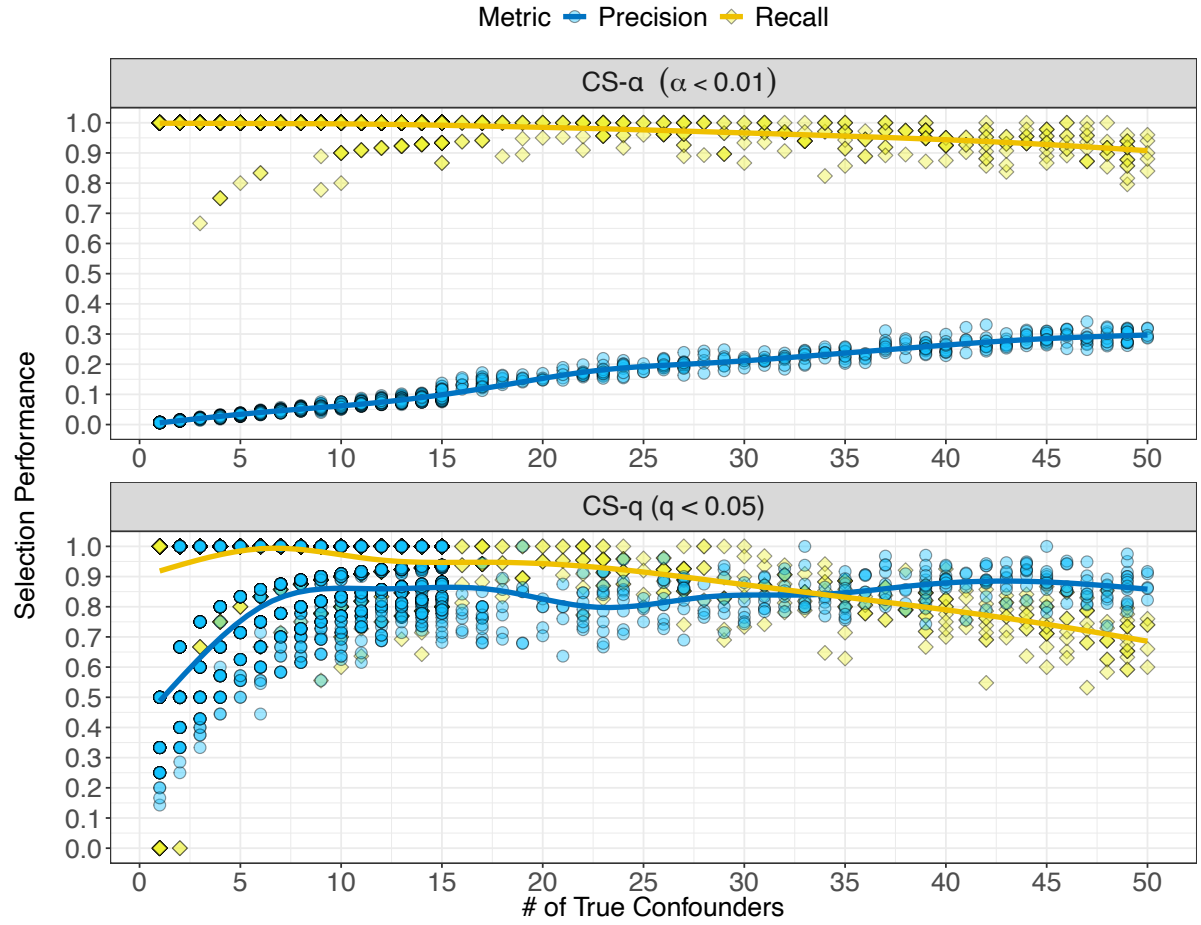


Figure 3: Precision and recall of confounder selection on simulated data by two multiple-testing correction methods: CS-q and CS- α . Each shape is a simulated data set. Shades are caused by overlapping shapes.

structure, and fully connected models than MRPC did. All these models contain a $T_1 - T_2$ edge, indicating the sensitivity of MRGN in detecting the $T_1 - T_2$ edge. On the other hand, compared to MRPC, this sensitivity led to reduced power to detect the null and conditional independence models, which have no $T_1 - T_2$ edge. MRGN also achieved comparable or better precision across nearly all models, demonstrating robustness to the high false positive rate of CS- α .

Table 2: Comparison of MRGN, MRPC and GMAC on simulated trios with confounders.

The oracle performance of MRGN using true confounders is shown to illustrate the upper bound due to noise in the data. For the other columns, the highest precision and recall in each row is highlighted. CS-q: confounder selection with the q -value method for FDR control at $q < 0.05$. CS- α : confounder selection with the type I error rate set at $\alpha < 0.01$ and no FDR control. MRPC cannot handle too many variables in reasonable runtime; we therefore did not run MRPC with the many confounders identified from CS- α . Rows 1-5 show model type-based performance, row 6 for all three edges in a trio, and the last row for only the $T_1 - T_2$ edge. * indicates that similar values are produced by the GMAC method.

	MRGN (True Confs.)		MRGN (CS-q)		MRGN (CS- α)		MRPC (CS-q)	
	Recall	Prec.	Recall	Prec.	Recall	Prec.	Recall	Prec.
Null Model (M0)	0.562	0.995	0.452	1.000	0.477	0.951	0.836	0.532
Mediation (M1)	0.982	0.516	0.877	0.438	0.883	0.429	0.097	0.147
V-structure (M2)	0.919	0.985	0.892	0.925	0.842	0.968	0.377	0.443
Cond. Indep. (M3)	0.515	0.989	0.283	0.910	0.305	0.872	0.609	0.828
Fully Connect. (M4)	0.600	0.566	0.501	0.425	0.464	0.419	0.217	0.592
All Edges	0.945	0.893	0.931	0.864	0.911	0.857	0.770	0.892
$T_1 - T_2$	1.000	0.767	0.994*	0.703*	0.9787	0.7071	0.7884	0.916

At the edge level, MRGN substantially outperformed MRPC and performed similarly to GMAC in recovering the $T_1 - T_2$ edge while maintaining high precision (Supplementary Figure 1). Because the GMAC mediation test is a special case of MRGN, the comparable performance indicates that GMAC’s notion of mediation encompasses multiple underlying causal

models. Recall and precision remained high for all edges except when the number of confounders exceeded 35, where recall declined but remained higher than that of MRPC (Supplementary Figure 2).

The impact of confounder selection on causal network inference

When we examined how confounder selection affected causal inference performance, we found no clear relationship between the number of falsely included confounders and MRGN’s ability to accurately infer causal edges (Figure 4a). In contrast, falsely omitted confounders had a strong negative impact on causal inference performance (Figure 4b).

We further observed substantially higher actual type I error rates for the $T_1 - T_2$ edge under the GMAC method than those reported by [16] (Supplementary Table 1). Their actual type I error rates were consistent with the significance level, such as $\alpha = 0.01$ or 0.05. However, the actual type I error rates in our simulation using GMAC are 0.73 at $\alpha = 0.05$ and 0.63 at $\alpha = 0.01$. Using MRGN, our actual type I error rates are also around 0.6 (with $\alpha = 0.01$ in causal network inference). We attribute these differences to several distinctions between our simulation framework and theirs.

We attribute these differences primarily to our more challenging simulation design. Compared with [16], we simulated substantially more confounders (between 1 and 50 per trio), aggregated them into a candidate pool exceeding 10,000 variables, and additionally included common-child and intermediate variables. In contrast, their simulations used a pool of 350 confounders (approximately two per trio on average) and considered each type of confounding separately.

To verify this explanation, we repeated the simulations of [16] using 2-15 confounders per trio while aggregating all confounders into a common candidate pool. The observed type I error rate matched the published results with two confounders, but increased steadily with the number of confounders and candidate pool size (Supplementary Figure 3), demonstrating that

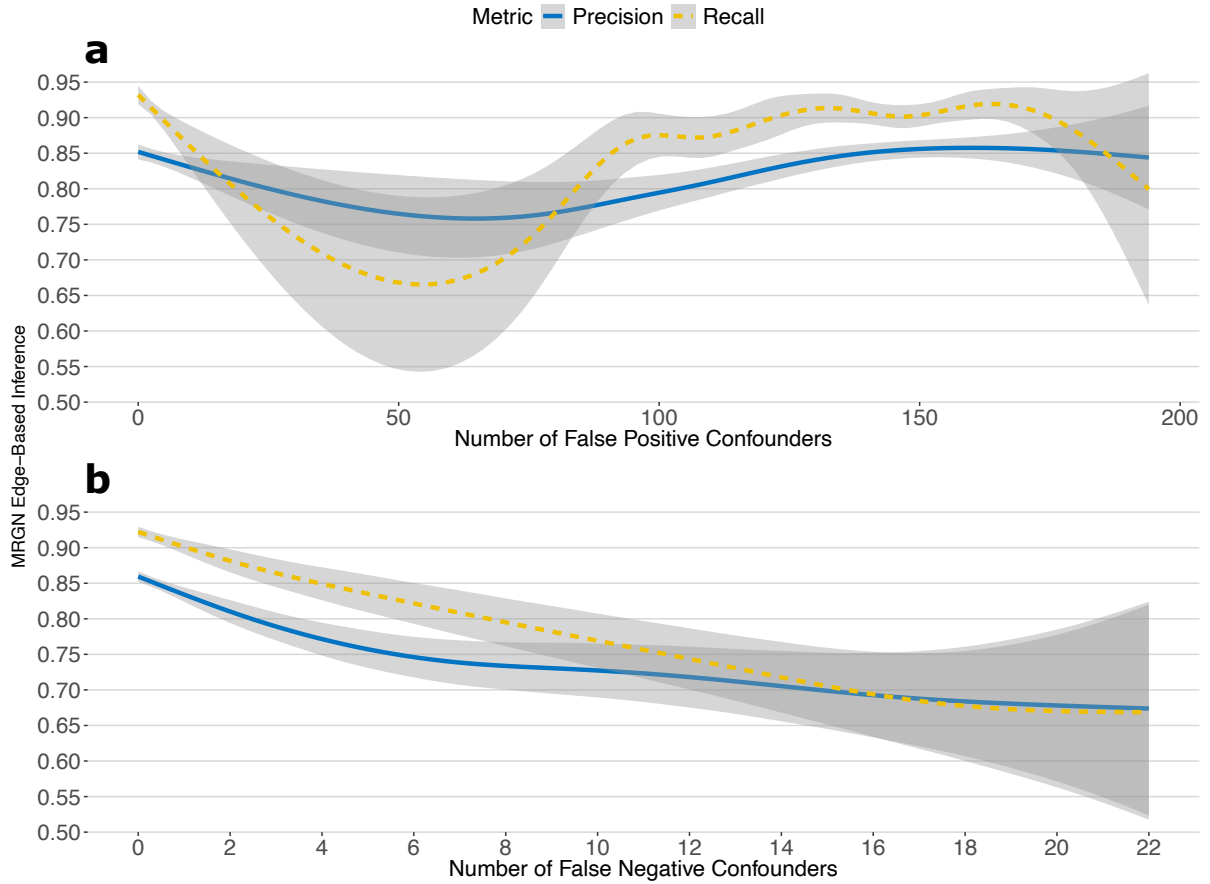


Figure 4: Relationship between confounder selection and causal inference performance measured as precision and recall of edge prediction. (a) Edge-based performance of MRGN on simulated trios as a function of the number of falsely included confounders. (b) Edge-based performance of MRGN on simulated trios as a function of the number of true confounders falsely omitted from the analysis.

Table 3: Inference times for MRGN, MRPC, and GMAC on simulated trios. We used the ADDIS method for FDR control in MRPC [13]. MRGN shows a substantially faster median inference time per trio compared to both MRPC-ADDIS and GMAC, demonstrating its computational efficiency in the presence of confounding.

Method	Median Time (s)	IQR (s)
MRGN	0.011	0.005
MRPC-ADDIS	0.460	9.012
GMAC	1.537	1.483

increasing confounding substantially inflates false positive $T_1 - T_2$ edges and reduces power to detect the null and conditional independence models.

Runtime comparison

Since MRGN consists of confounder selection and causal network inference, we report the computation time for each step separately. We evaluated the runtime of confounder selection using confounder pools of 100, 500, and 1000 covariates and 500 or 1000 trios, yielding six scenarios. In the most computationally intensive setting (1000 covariates and 1000 trios), MRGN completed confounder selection in 16 minutes (Figure 5), compared with approximately 28 minutes by [16]. This improvement reflects the more efficient confounder selection procedure in MRGN.

For the step of causal network inference, we evaluated the runtime on all the 1,800 trios. MRGN is substantially faster than both MRPC and GMAC (Table 3). The median time to analyze a trio using the same set of confounders was 0.0065 seconds for MRGN, compared to 0.3557 seconds for MRPC and 0.7088 seconds for GMAC. This makes MRGN over 50 times faster than MRPC and more than 100 times faster than GMAC, on average. For GMAC, the reported runtimes are based on the default single-threaded implementation provided in its R package; enabling its parallelization option could further reduce its computation time.

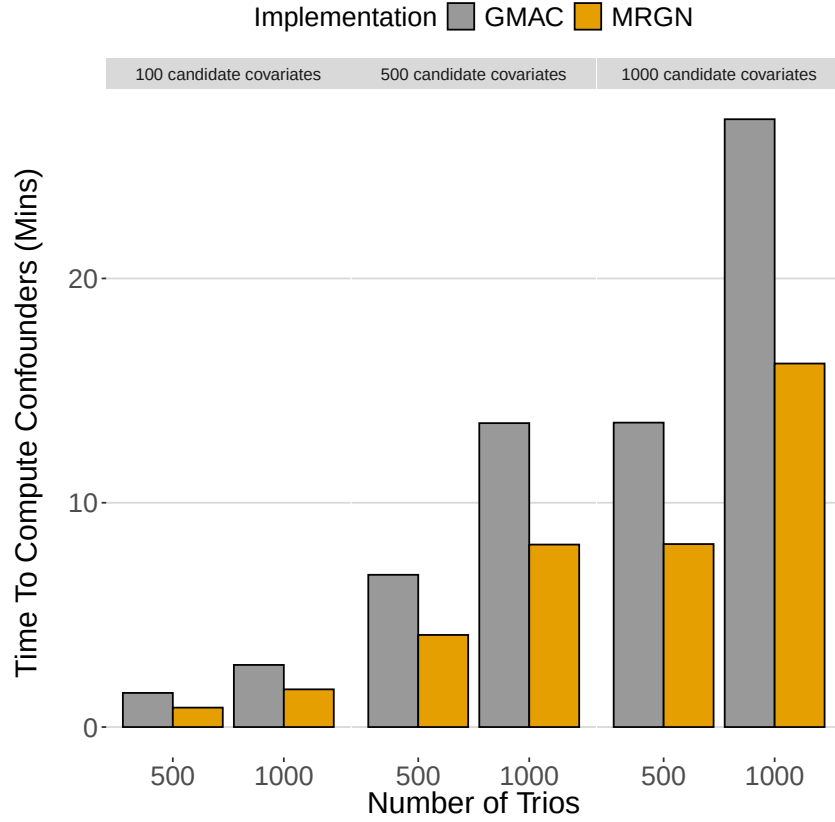


Figure 5: Comparison of computation times between our implementation of the CS-q method for confounder selection and that of the GMAC method [16]. Computation times are shown across different numbers of input trios and sizes of the candidate confounder pool.

3.2 Application to GTEx data

The GTEx consortium has mapped a large number of genetic variants as expression quantitative trait loci (eQTLs), which are associated the expression of most genes in the human genome across many human tissue types [25]. Recent studies have focused on identifying cis-target genes that mediate the relationship between the eQTL and its associated trans-target genes [16]. We previously have applied the MRPC method to all the GTEx tissues to detect diverse causal models for such trios: an eQTL and its cis- and trans-target genes, where the genes are protein coding genes or long non-coding RNAs (lncRNAs) [13].

We applied the preprocessing and normalization pipeline of [13] to whole blood data from 670 GTEx v8 samples. Gene expression was PEER-normalized [28]. Each trio consisted of an eQTL and its cis- and trans-target genes of highest association, where cis-genes were located within 1 Mb of the eQTL and trans-genes were located more than 1 Mb away or on a different chromosome. After preprocessing, 8,823 trios were identified. Restricting the analysis to protein-coding genes and long non-coding RNAs (lncRNAs) to reduce transcript mismapping yielded a final set of 3,248 trios (Supplementary Table 2). Because GMAC cannot accommodate missing genotype values, we imputed missing genotypes using multiple correspondence analysis implemented in the *missMDA* R package [29].

We derived 670 principal components (PCs) from genome-wide PEER-normalized expression data as candidate confounders. Using CS- α with $\alpha < 0.05$, MRGN selected a median of 51 confounders per trio (SD = 6.38) (Figure 6a). Sex, platform, and PCR batch were also included.

MRGN identified more mediation and conditional independence trios

MRGN identified 911 trios as the null model, 91 as mediation, 10 as v-structures, 2,062 as conditionally independent, 104 as fully connected, and 70 as “other” (Figure 6b). Although the total number of inferred $T_1 - T_2$ edges was similar across methods (Figure 6c), MRGN iden-

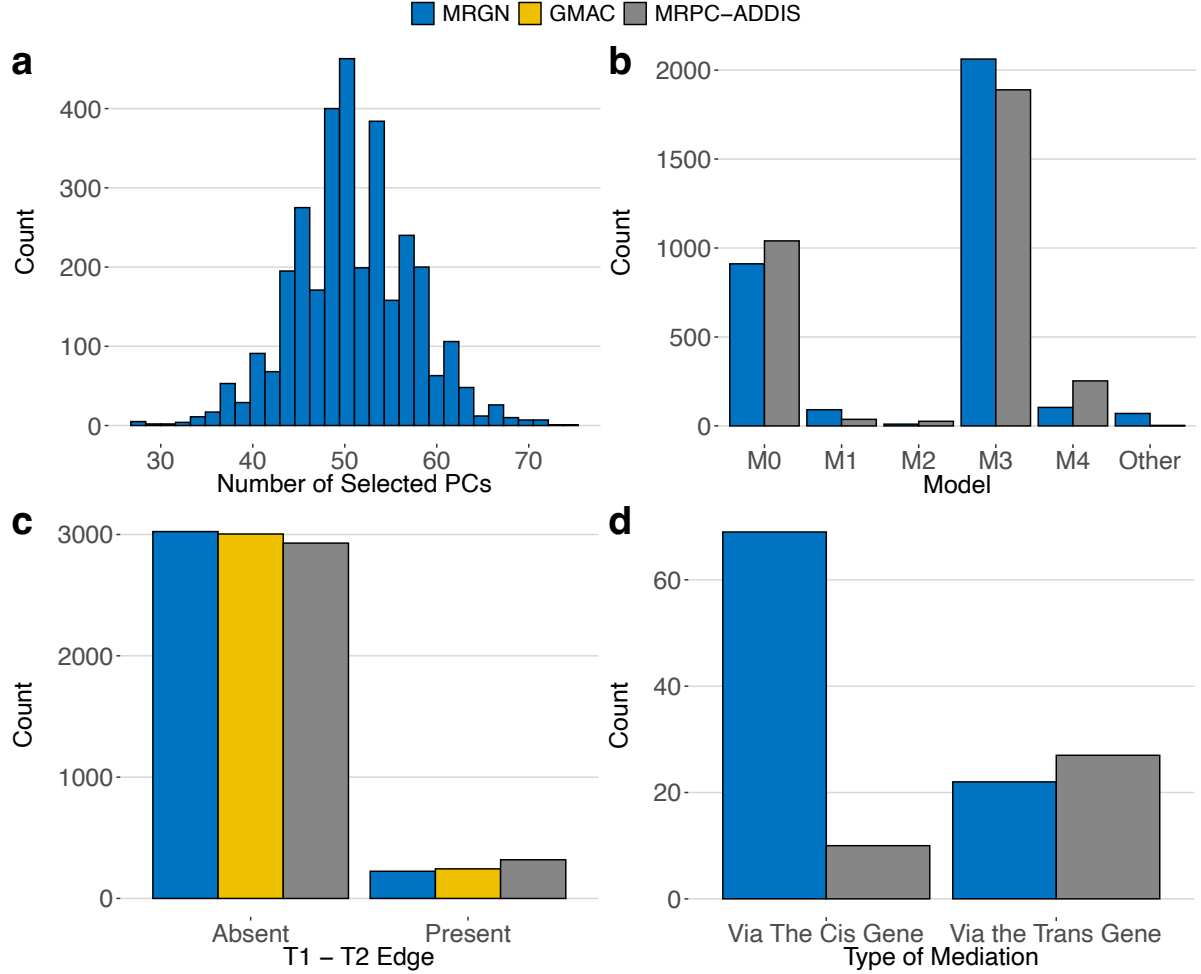


Figure 6: Comparative analysis of MRGN, MRPC [30] and GMAC [16] on GTEx human whole blood trios. We applied MRGN, GMAC, and MRPC (with ADDIS for FDR control) to genomic trios. (a) Histogram of the number of Principal Components (PCs) selected as confounders per trio. (b) Count of inferred causal models (M1–M4 and Other) comparing MRGN and MRPC. (c) Counts of trios with the $T_1 - T_2$ edge inferred as absent or present by the three methods. (d) Counts of two types of mediation models (via cis-gene vs. trans-gene) inferred by MRGN and GMAC.

308 tified more than twice as many mediation trios and 173 additional conditional independence
309 trios than MRPC [13]. As in MRPC, MRGN distinguished two mediation models (Figure 6d):
310 cis-gene mediation ($V \rightarrow T_1 \rightarrow T_2$) and trans-gene mediation ($V \rightarrow T_2 \rightarrow T_1$). Among the 91
311 mediation trios, 69 (75.8%) involved cis-gene mediation and 22 (24.2%) involved trans-gene
312 mediation (Supplementary Table 3).

313 Only eight cis-gene mediation trios and six trans-gene mediation trios were identified by
314 both methods, suggesting substantial differences in the inferred causal models. Examination of
315 trios classified as mediation by MRGN but not MRPC indicated that these discrepancies were
316 primarily driven by differences in confounder selection. For example, for the trio consisting of
317 rs117667029, METTL26, and PTMS, MRGN identified 49 confounders and inferred media-
318 tion ($\text{rs117667029} \rightarrow \text{METTL26} \rightarrow \text{PTMS}$), whereas MRPC identified only three confounders
319 and inferred conditional independence ($\text{METTL26} \leftarrow \text{rs117667029} \rightarrow \text{PTMS}$). Repeating the
320 MRGN analysis with the same three confounders also yielded conditional independence, con-
321 firming that accurate confounder selection is critical for consistent causal network inference.

322 **MRGN is robust to model misspecification in inference of mediation**

323 We further investigated the effect of inference model misspecification on mediation inference
324 using 192 mediation trios identified by GMAC in whole blood. Four scenarios were considered,
325 ranging from models using the true confounders to models omitting many moderate-to-strong
326 confounders (see Supplementary Text 1).

327 Recall remained around 93% when the inference model included most influential con-
328 founders, even if a few weak confounders were incorrectly included or omitted. In contrast,
329 omitting many moderate-to-strong confounders reduced recall to approximately 70%, demon-
330 strating that retaining influential confounders is critical for accurate causal network inference.

4 Discussion

In summary, we have developed MRGN to improve the power and accuracy of detecting key causal relationships in diverse genomic trios, particularly in the presence of multiple confounding variables. Our method adopts an extended interpretation of the PMR [7] and prioritizes conditional independence tests to identify five basic causal trio structures. A major advantage of our approach is the reduction of the multiple testing burden; by adjusting for selected confounders within a regression framework, MRGN can incorporate a large number of confounders without the need to infer a graph larger than three nodes.

Our simulation results further underscore the critical importance of confounder selection in causal inference. Omitting key confounders drastically reduces the power to detect mediation, leading to Type II errors and the loss of regulatory insights. Conversely, the inclusion of false-positive confounders has a negligible impact, particularly when their effects are weak. This suggests that in practical applications, a liberal strategy favoring the inclusion of many potential confounders is preferable. Ultimately, MRGN balances this requirement without excessive computational cost, making it a robust tool for causal network inference. However, when the sample size is not much larger than the number of identified confounders, including many false confounders can substantially increase the number of regression coefficients, leading to unreliable estimation. We are currently developing extensions of MRGN to address issues with a small sample size.

Constraint-based methods such as MRPC initialize learning by filtering edges from a fully connected graph structure. After removing edges with weak marginal relationships, the remaining edges are tested sequentially by conditioning on an increasing neighborhood of nodes [7, 31]. Critically, MRPC treats confounding variables as additional nodes within the graph. When confounders are included, the number of conditional tests increases significantly, and only edges with strong supporting evidence survive. Thus, MRPC inherently infers a larger

network than MRGN for the same biological question. Additionally, the adaptive FDR can result in significance thresholds that deviate substantially from the fixed $\alpha = 0.01$ employed by MRGN. As demonstrated by our simulations, this causes MRPC to act more conservatively in the presence of confounding variables.

While our simulations and real data analysis demonstrate the flexibility of MRGN in robustly estimating causal relationships, the method is currently limited to trios containing a genetic variant that acts as an instrumental variable. Similar to MRPC, MRGN relaxes strict instrumental variable assumptions by empirically validating the association between the variant and the phenotype(s). We are currently extending this approach to develop a general MRGN algorithm capable of accommodating larger networks and other types of confounding variables.

Data Availability

Data from The Genotype-Tissue Expression (GTEx) Project version 8 were obtained from dbGaP (<http://www.ncbi.nlm.nih.gov/gap>) through the following accession number: phs000424.v8.pht002741.v8.p2. Open-source R packages used in this manuscript include MRGN; <https://github.com/audreyfulab/MRGN>, MRPC (version 3.1.0); <https://cran.r-project.org/web/packages/MRPC/index.html>, and GMAC (version 3.1); <https://cran.r-project.org/web/packages/GMAC/index.html>. R code for simulations and applications with results is available at https://github.com/audreyfulab/MRGN_Manuscript_Repo. Additional results are also included in the supplementary material.

Conflicts of Interest

The authors declare no competing interests.

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Author Contributions

J.K and A.Q.F designed the study, J.K developed the tools and analyzed the data, and J.K and A.Q.F wrote the manuscript.

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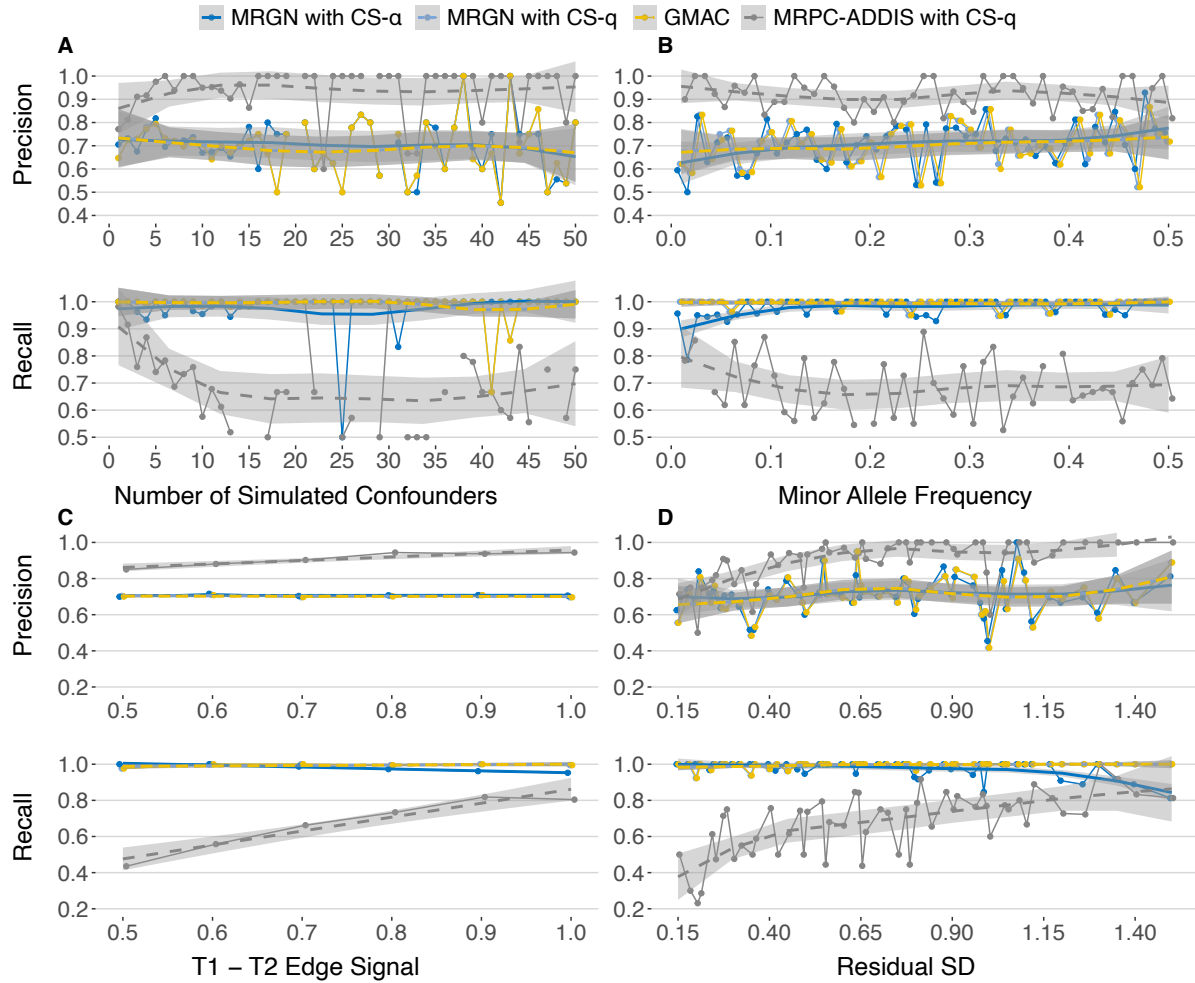
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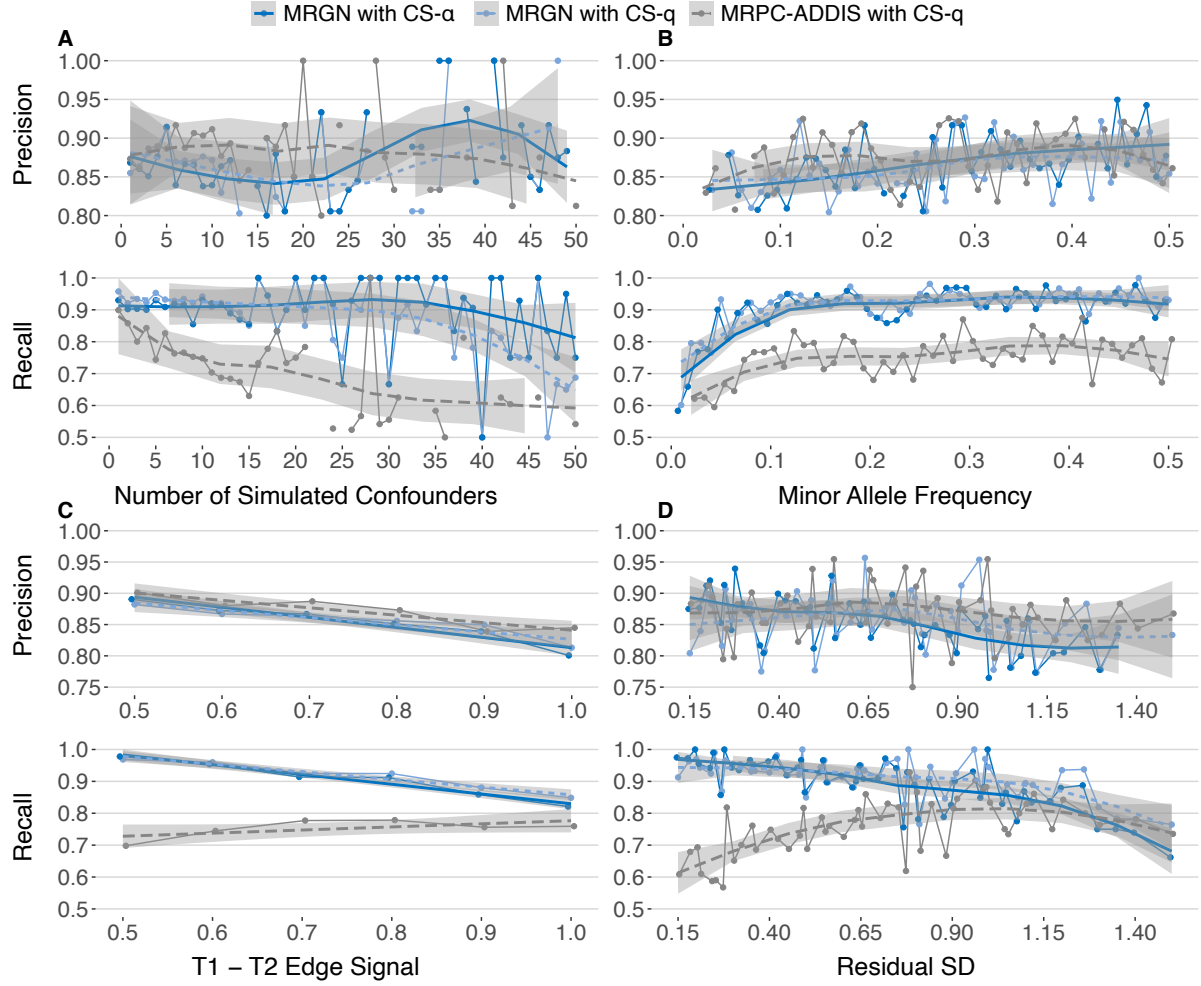
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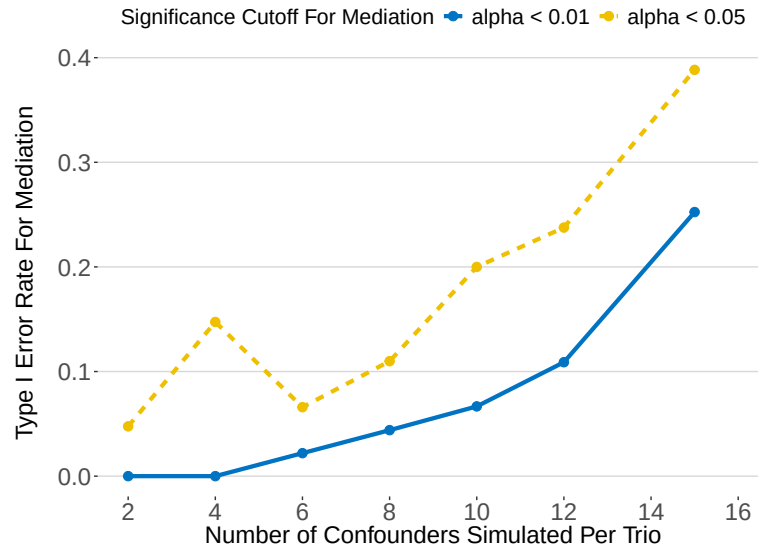
Supplementary figures



Supplementary Figure 1: Precision-recall curves for predicting the $T_1 - T_2$ edge across multiple simulation parameters. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



Supplementary Figure 2: Precision and recall curves of edge-based prediction performance for MRGN, and MRPC with confounder selection. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



Supplementary Figure 3: Comparison of GMAC and MRGN type I error rates for $T_1 - T_2$ detection under confounding. Results of simulations investigating type I error rate (false mediation) under the null model. Trios were simulated under the null model for each of seven scenarios: 2, 4, 6, 8, 10, 12 or 15 confounders, excluding common child and intermediate variables. The GMAC method was applied to trios in each scenario using CS-q confounder selection.

498 Supplementary tables

Supplementary Table 1: Inference for the $T_1 - T_2$ edge in simulated trios by MRGN, MRPC, and GMAC. The table summarizes the number of trios inferred to detect (or miss) the presence (or absence) the $T_1 - T_2$ edge, as well as recall, precision, and actual type I and II error rates. Different confounder selections were applied in MRGN and GMAC.

Supplementary Table 2: Inference results from MRGN, MRPC, and GMAC for 3,248 whole blood trios from GTEx v8. All analyzed GTEx whole-blood trios, including genetic annotations (locus, variant ID, gene name, etc.) and inference results from MRGN, MRPC-ADDIS, and GMAC.

Supplementary Table 3: Comparison of MRGN, MRPC, and GMAC on GTEx whole blood trios. CS- α confounder selection was used in MRGN.

499 **Supplementary text**

Supplementary Text 1: Details of the four simulation scenarios in section “MRGN is robust to model misspecification for inference of mediation”.